**Solution**

**ELECTRICITY -04**

**Class 10 - Science**

1. This is because when the bulb is switched on, then the temperature of filament rises, and we know that resistance is directly proportional to the temperature. That's why resistance of filament increases.
2. His observations are $I_1 < I_2$ because, in a series combination resistance is higher, so the current is less. But $V_1 = V_2$ because in both the cases battery is the same. (For parallel combination current will be more.)
3. Voltmeter is always connected in parallel to the ends of the resistor across which the potential difference is required to be measured.
4. The relation is given by H =VIt.
   Here H=Heat produced, V=Potential difference, I=Current produced and t=time.
5. SI unit of resistivity is ohm-m (Ω·m) .
6. The electrical work done per unit time is called electric power.
7. Parallel combination
8. Resistance is equal to the slope of V-I graph. Here, slope of graph for temperature $T_2$ is higher, so resistance for temperature $T_2$ is higher, As resistance is directly proportional to the temperature.
9. Rubber is an electrical insulator. Hence electrician can work safely while working on an electric circuit without a risk of getting any electric shock.
10. A closed path is required for the flow of current so that charges can move in a particular direction in a given circuit. But if path is not closed means circuit is open, then there is air between the gap, and we know that air is an insulator for flow of charges, so the current stop flowing in the circuit.
11. Current always flows from a higher potential to a lower potential end of the conductor. So end 'A' of the conductor is at a higher potential.
12. Electrons flow from negative to positive i.e. in the direction opposite to that of conventional current.
13. The practical unit of power is horse power (H.P.).
    1 H.P. = 746 watt.
14. i. Zero
    ii. Infinite
15. ammeter in series and voltmeter in parallel.
16. Let the three resistors are $R_1, R_2$ and $R_3$. Here $R_1$ and $R_2$ are parallel to each other and $R_3$ is in series with them then equivalent resistance can be obtained by the given formula:
    $\therefore \quad \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{6} + \frac{1}{2} = \frac{1+3}{6} = \frac{4}{6} = \frac{2}{3}$
    $\Rightarrow \quad R = \frac{3}{2}\Omega$
    Now, R and $R_3$ are in series.
    $\therefore$ Combind resistance $R_4 = R + R_3$
    $= 3/2 + 6 = 1.5 + 6 = 7.5 Ohm$
17. For getting minimum resistance R we can connect five resistors in parallel Combination .We know that :$R = 1/5, n = 5$
    $R_{\text{eq}} = \frac{R}{n} = \frac{1/5}{5} = \frac{1}{25}\Omega$
18. 1kWh = 1,000 Ω × 1h = 1,000 J/S × 3600S
    1kWh = 3,600,000 J = $3.6 \times 10^6$ J
19. a. P = VI
    b. $P = I^2R$
    c. $P = \frac{V^2}{R}$
20. When a wire of resistivity rho is pulled to double it's length then new resistivity of conducting wire will not change as resistivity depends on the nature of material not on the length of conductor.
21. It is connected in series so that whole of electric current, which it has to measure, passes through it.
22. Electrical resistivity of a material of a conductor is the resistance offered by unit length and unit area of cross section of a wire of the material.
    Its S.I. unit is ohm-metre.

23. Here, charge (Q) = 1500 C, time (t) = 2 min 30 s = (2 $\times$ 60 + 30) s = 150 s

$\therefore$ Current (I) = $\frac{Q}{t}$

$= \frac{1500\mathrm{C}}{150\,\mathrm{s}}$ = 10 A

24. Resistance.

25. Nichrome, due to its high resistivity.

26. We are given, potential difference V = 60 V, current I = 4 A.

According to Ohm's law, $R = \frac{V}{I} = \frac{60\mathrm{V}}{4\mathrm{A}} = 15\Omega$

When the potential difference is increased to 120 V the current is given by

Current $= \frac{V}{R} = \frac{120\mathrm{V}}{15\Omega} = 8\mathrm{A}$

The current through the heater becomes 8 A if the potential difference is increased to 120 V.

27. We know that resistance and cross-section area are inversly proportional to each other. So less area of cross section means more resistance .

R $\propto \frac{1}{A}$ [since lengths are same]

So, out of two, wire Q has greater resistance.

28. We know that, $I = \frac{ne}{t}$

$\Rightarrow n = \frac{It}{e}$

Here I = 1 A, t = 1 s and e = 1.6 $\times$ $10^{-19}$ C

$\therefore$ Number of electrons, n = $\frac{\mathrm{I}t}{e}$

$= \frac{1\times1}{1.6\times10^{-19}}$

= 6.25 $\times$ $10^{18}$

29. P = VI

= 220V $\times$ 0.50 A

= 110 J/s

= 110 W

The power of the bulb is 110W.

30. We are given, I = 0.5 A; t = 10 min = 600 s.

we have,

Q = I $\times$ t

= 0.5 A $\times$ 600 s

= 300 C

Hence, the amount of electric charge that flows through the circuit is 300 C

31. While glowing, I = 450 mA = 0.45 A, V = 6 volt

Using Ohm's law, Resistance of bulb, $\mathrm{R} = \frac{\mathrm{V}}{\mathrm{I}} = \frac{6}{0.45} = 13.33\Omega$

The reason for the difference in resistance of bulb when cold (R = 2.5 Ω) and while glowing (R = 13.33 Ω), is that the resistance of filament of bulb increases with the increase in temperature.

32. The total energy consumed by the refrigerator in 30 days would be

400 w $\times$ 8.0 hour/day $\times$ 30 days = 96000 W h

= 96 kWh

Thus the cost of energy to operate the refrigerator for 30 days is

96 kW h $\times$ ₹ 3.00 per kWh = ₹ 288.00

33. When the equivalent resistance of connecting wire is low then, wire should be connected in parallel combination. So equivalent resistance can be obtained by the given formula :

$\therefore \frac{1}{R_{\mathrm{eff}}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} = \frac{1}{2}$

$\Rightarrow \quad R_{\mathrm{eff}} = 2\Omega$

34. $R_s$ = 0.2 + 0.3 + 0.4 + 0.5 + 12 = 13.4Ω

I = $\frac{V}{R} = \frac{9}{13.4} = \frac{90}{134}$ or or I = 0.6716A

When in series, the current passing through each resistor is equal to the current in the circuit.

$\therefore$ Current through 12Ω = 0.6716 A

35. The current flowing through the conductor is reduced to one-half of its previous value in accordance with the relation I = $\frac{V}{R}$.

36. The potential difference between two points A and B is said to be one volt if 1 joule of work is done to move 1 coulomb of charge from one point to another point in an electric field.

37. a. Resistance of appliance = $\frac{(\text{Voltage rating of appliance})^2}{(\text{Power rating of appliance})}$

$R = \frac{V^2}{P}$

b. Safe current = $\frac{\text{Power rating of appliance}}{\text{Voltage rating of appliance}}$

$I = \frac{P}{V}$

38. $IR^2$ does not represent electrical power in the circuit.
39. Electrons are flowing from higher potential end to lower potential end through the conductor.
40. The potential difference (voltage) remains constant.
41. The obstruction offered to the flow of current by a conductor is called its resistance.
42. 2$\Omega$ resistances are in parallel. So Now, all three resistances in series. So

$R_{eq} = 1 + 1 + 6 = 8\Omega$

$I = \frac{V}{R}$

$= \frac{4}{8}$

= 0.5 Amp

43. R = 20 $\Omega$; I = 5A; t = 30 s.

$H = I^2Rt = (5)^2\ (20)\ (30)$

H = 15,000 J

44. The potential difference between two points is said to be 1 volt if 1 joule work is done in bringing 1 coulomb charge from one point to the other.

1 volt = $\frac{1 \text{ joule}}{1 \text{ coulomb}}$

45. Here, l = 150 cm; a = 0.015 $cm^2$; R = 3.0 $\Omega$

Specific resistance, $\rho = \frac{Ra}{l}$

$= \frac{3.0 \times 0.015}{150}$

= 0.0003 $\Omega$cm

46. Because the resistivity of an alloy is higher than the pure metal.
47. X = Ammeter, Y = Rheostat, Z = Voltmeter.
48. a. $R = R_1 + R_2 + R_3$

b. $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

49. Fuse is a safety device which is used in an electric circuit. Fuse is always connected in series. When the circuit gets more amount of current or it overloaded then fuse wire melts and current stop flowing through the circuit and save our appliances.
50. According to Ohm's law, the current flowing in a conductor is directly proportional to the potential difference applied across its ends, provided the temperature and other physical conditions of the conductor remain constant.
51. If anyone stops working due to some reason, other will also stop working.
52. Heat produced, $H = I^2Rt$.
53. If one bulb blows off in a series circuit then all other bulbs stop glowing because current stop flowing .
54. The flow of electric charges across a cross-section of a conductor constitutes an electric current. For example, a stream of electrons moving through a conducting wire constitutes an electric current.
55. For filament of electric lamp we require a strong metal with high melting point. Tungsten is used exclusively for filament of electric lamps because its melting point is extremely high (3410$^{o}$C)
56. Here, I = 5.0 A, R = 44 $\Omega$, V = ?

Using Ohm's law, V = IR

= 5.0 $\times$ 44 = 220V

57. The amount of charge Q, that flows between two points at potential difference V (= 12 V) is 2 C. Thus, the amount of work W, done in moving the charge [from Eq. ($V = \frac{W}{Q}$ )] is

W = VQ

= 12 V $\times$ 2 C

= 24 J

Hence, 24 J of work is done in moving a charge of 2 C across two points having a potential difference 12 V.

58. A galvanometer is an instrument which can detect the presence of electric current in a circuit.
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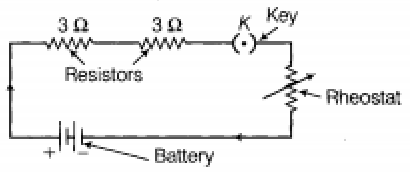
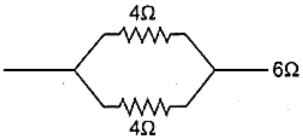
60. Resistance of a conductor is inversely proportional to its area cross-section. As resistivity increases area decreases and as area increases, resistivity decrease.
61. Current = $\frac{\text{Charge}}{\text{Time}}$ or I = $\frac{Q}{t}$
62. A superconductor is a substance of zero resistance at very low temperatures.
    **Example:** Mercury below 4.2 K, Lead below 7.25 K.
63. An electric circuit is a continuous and closed path of an electric current. If the electric circuit is complete, electric current can flow through the circuit. If the circuit is broken anywhere or switch of the circuit is turned off,the current stops flowing.
64. The substances through which electricity can flow are called conductors and the substances through which electricity cannot flow are called insulators.
65. When we connect rheostat, key, battery of two cells and two resistors of three ohm each then required circuit diagram will be:




66. We know that, $V = \frac{W}{Q}$
    Therefore, the amount of work done in moving 1.5 C charge across a p.d. of 4 V is,
    W = VQ
    = 4 × 1.5 J
    = 6 J
67. The change in the current flowing through the electrical component can be determined by Ohm's Law.
    According to Ohm's law i.e.,
    V = IR
    V: applied voltage
    I: current
    R: resistance
    $\Rightarrow I = \frac{V}{R}$
    Now potential difference becomes one fourth,
    $V' = \frac{V}{4}$
    So current will be: $I' = \frac{(\frac{V}{4})}{R}$
    $\Rightarrow I' = \frac{v}{4R}$
    $\Rightarrow \mathrm{I}' = \frac{\mathrm{I}}{4}$
    When the potential difference is one fourth, the current through the component also decreases to one fourth of its initial value.
68. Ammeter is a apparatus to measure electric current in a circuit.
69. The resistivity of a substance is numerically equal to the resistance of a rod of that substance which is 1 metre long and 1 square metre in cross-section.
70. Silver, because its electrical resistivity is least out of the given materials.
71. Unit of Resistance : R = $\frac{V}{I}$
    If V = 1 volt; I = 1 ampere then R = 1 ohm
    i.e. the resistance of a substance is 1 ohm (or 1 Ω) if 1 A of the current flows through it when a potential difference IV is applied across its ends.
72. Given energy =2 Kwh, power=100 W
    We also know that:
    Time, t = $\frac{\text{Energy}}{\text{Power}} = \frac{\text{2kWh}}{\text{100W}}$
    $= \frac{\text{2000Wh}}{\text{100W}}$ = 20h
    So time taken by the bulb to consume 2kwh energy =20 hours.
73. Resistance of each part, $R_1 = \frac{32}{4}\Omega = 8\Omega$
    When connected in parallel, the equivalent resistance is
    $\frac{1}{R} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$
    or R = 2Ω

74. 


In parellel,

$\frac{1}{R'} = \frac{1}{4} + \frac{1}{4}$

R' = $\frac{4}{2}$ = 2Ω

In series, R = R' + R"

2Ω + 6Ω = 8Ω

75. SI unit of electric current is Ampere. Current is said to be 1 ampere (1 A), if 1 coulomb charge flows per second across a cross-section of a conductor.